UNIVERSITY MEDICAL CENTER

Department of Cardiology – Inpatient Consultation Service

1500 East Medical Center Drive • Springfield, MA 01199

Phone: (413) 555-7200 • Fax: (413) 555-7209

INPATIENT CARDIOLOGY CONSULTATION

Patient Name:	Brady998 Hickle134
Date of Birth:	November 27, 1914
Age:	81 years
Sex:	Male
Medical Record #:	MRN-784521-9
Consultation Date:	August 9, 1996
Consulting Physician:	Richard T. Morrison, MD, FACC
Requesting Physician:	Dr. Angela M. Patterson, Intensive Care Unit
Location:	Medical ICU, Bed 7

REASON FOR CONSULTATION

Cardiac arrest survivor. Evaluation and management recommendations for post-arrest care, risk stratification, and long-term arrhythmia management.

HISTORY OF PRESENT ILLNESS

Mr. Hickle134 is an 81-year-old gentleman with extensive cardiac history including coronary heart disease with prior percutaneous coronary intervention, type 2 diabetes mellitus, hypertension, prior stroke with residual left arm monoparesis, and current treatment for prostate carcinoma. He experienced a witnessed out-of-hospital cardiac arrest earlier today, August 9, 1996, at approximately 0730 hours.

According to emergency medical services (EMS) report and collateral history from family, the patient collapsed suddenly at home while preparing breakfast. His wife witnessed the event and immediately called 911. She reports he became unresponsive and was not breathing normally. A neighbor who is a nurse initiated bystander CPR within approximately two minutes of collapse. EMS arrived approximately six minutes after the initial call.

On EMS arrival, the patient was found to be in ventricular fibrillation. Defibrillation was delivered with 200 joules biphasic, CPR was continued, and intravenous epinephrine 1 mg was administered. After approximately 8 minutes of resuscitation, return of spontaneous circulation (ROSC) was achieved in the field following the second defibrillation attempt and administration of amiodarone 300 mg IV bolus. Total downtime was estimated at 10-12 minutes.

The patient was intubated in the field for airway protection given post-arrest altered mental status. He was transported emergently to University Medical Center Emergency Department, arriving at approximately 0815 hours. On arrival, he had palpable pulses, blood pressure 98/62 mmHg on norepinephrine infusion, heart rate 88 bpm, and was unresponsive (GCS 3T). He was admitted to the Medical ICU and targeted temperature management protocol was initiated with goal core temperature of 33°C for 24 hours.

Initial laboratory studies showed troponin I elevated at 2.8 ng/mL, BNP 1240 pg/mL, potassium 3.6 mmol/L, magnesium 1.9 mg/dL, creatinine 1.8 mg/dL (baseline 1.5-1.6), and lactate 6.2 mmol/L. Arterial blood gas on 100% FiO_2 revealed pH 7.28, pCO_2 42, pO_2 198, HCO_3 19.

PAST MEDICAL HISTORY

- **Coronary Heart Disease:** Known multivessel disease. Prior percutaneous coronary intervention with stent placement (specific details being obtained from outside records). History of anteroseptal myocardial infarction. Chronic stable angina managed medically.
- **Hypertension:** Long-standing, on multiple agents.
- **Diabetes Mellitus, Type 2:** With associated diabetic nephropathy (CKD Stage 1) and hyperglycemia requiring insulin therapy.
- **Cerebrovascular Disease:** Prior ischemic stroke with residual left arm monoparesis and abnormal gait. History of silent cerebral microhemorrhages on prior MRI.
- **Obesity:** BMI >30.
- **Metabolic Syndrome:** With associated hypertriglyceridemia and prediabetes.
- **Prostate Carcinoma:** Carcinoma in situ, currently on hormonal therapy with leuprolide and chemotherapy with docetaxel.
- **Anemia:** Chronic, on epoetin alfa and iron supplementation.
- **Chronic Kidney Disease Stage 1:** Secondary to diabetic nephropathy. Has required intermittent hemodialysis.
- **Osteoarthritis:** Bilateral hips.
- **Alcoholism:** Remote history.
- **Chronic Sinusitis:** Currently with acute viral pharyngitis and sinusitis.
- **Gastrointestinal:** History of colonic polyps status-post polypectomy.
- **Concussion:** Remote, without loss of consciousness.

Prior Cardiac Procedures:

- Biventricular implantable cardioverter-defibrillator (ICD) insertion (date and indication being clarified)
- Catheter ablation of cardiac tissue (specific substrate being clarified from records)
- Percutaneous coronary intervention with stent placement
- Multiple echocardiograms

CURRENT MEDICATIONS

(As documented prior to admission)

- Metformin HCl 500 mg extended-release oral tablet, once daily
- Amlodipine 5 mg oral tablet, once daily
- Hydrochlorothiazide 25 mg oral tablet, once daily
- Lisinopril 10 mg oral tablet, once daily
- Humulin 70/30 insulin suspension, subcutaneous twice daily
- Nitroglycerin 0.4 mg sublingual spray, as needed for chest pain
- Simvastatin 20 mg oral tablet, once daily at bedtime

- Naproxen sodium 220 mg oral tablet, as needed
- Docetaxel 20 mg/mL injection, per oncology protocol
- Leuprolide acetate 30 mg/mL injection, per oncology protocol
- Ferrous sulfate 325 mg oral tablet, once daily
- Epoetin alfa 4000 units injection, three times weekly
- Calcium carbonate 1250 mg / Vitamin D3 1000 units / Vitamin K 0.4 mg chewable tablet, once daily
- Amoxicillin-clavulanate 250/125 mg oral tablet, for current sinusitis

Resuscitation Medications Administered:

- Epinephrine 1 mg IV (in field)
- Amiodarone 300 mg IV bolus (in field), followed by 150 mg IV bolus in ED
- Atropine sulfate 1 mg IV (in ED)
- Alteplase 100 mg (consideration documented but not administered)

ALLERGIES

No known drug allergies.

SOCIAL HISTORY

Retired factory worker. Lives at home with wife of 58 years. Two adult children who are involved in his care. Remote history of alcohol use disorder, now abstinent for over 20 years. Former smoker, quit approximately 30 years ago (30 pack-year history). No illicit drug use.

FAMILY HISTORY

Father deceased from myocardial infarction at age 67. Mother deceased from stroke at age 72. One brother with coronary artery disease. No family history of sudden cardiac death or inherited arrhythmia syndromes.

PHYSICAL EXAMINATION

General:	Intubated, sedated male. Cooling protocol in place with surface cooling pads. Core temperature 34.2°C.
Vital Signs:	BP 102/58 mmHg (on norepinephrine 0.08 mcg/kg/min), HR 84 bpm, RR 14 (ventilator rate), SpO2 98% on FiO2 0.60.
HEENT:	Normocephalic. Pupils equal, sluggish response to light (sedation effect). Endotracheal tube in place, secured at 23 cm at the lip. Orogastric tube in place.
Neck:	Supple. JVP not elevated. No bruits appreciated.
Cardiovascular:	Regular rate and rhythm. Normal S1, S2. Soft II/VI systolic murmur heard best at apex, radiating to axilla, consistent with mitral regurgitation. No S3 or S4 appreciated. Distal pulses 2+ and symmetric in upper extremities, 1+ in lower extremities bilaterally.
Respiratory:	Bilateral breath sounds clear anteriorly. No wheezes, rales, or rhonchi appreciated on anterior examination. Decreased air entry at bases.
Abdomen:	

	Soft, non-distended. Bowel sounds hypoactive. No masses or organomegaly appreciated. No bruits.
Extremities:	Trace bilateral lower extremity edema. No cyanosis. Warm and well-perfused. Left upper extremity with decreased muscle tone consistent with prior stroke.
Neurological:	Sedated for therapeutic hypothermia protocol. No spontaneous movements. Corneal reflexes intact. Gag reflex present. Detailed neurological examination deferred pending completion of TTM protocol and neurological consultation.
Skin:	Cool to touch secondary to TTM. No rashes or lesions noted on exposed areas.

DIAGNOSTIC STUDIES

12-Lead Electrocardiogram (August 9, 1996, 0825 hours):

Sinus rhythm at 86 beats per minute. Normal axis. PR interval 168 ms. QRS duration 102 ms. QTc 448 ms. Poor R-wave progression V1-V3 with QS complexes consistent with old anteroseptal myocardial infarction. No acute ST-segment elevation or depression. T-wave flattening in lateral leads (I, aVL, V5-V6). No comparison EKG available at this time.

Chest Radiograph (August 9, 1996, Portable AP):

Endotracheal tube tip 4 cm above carina, satisfactory position. Orogastric tube courses below diaphragm. Cardiac silhouette moderately enlarged (cardiothoracic ratio approximately 0.58). Biventricular ICD with right atrial, right ventricular, and coronary sinus leads in standard positions; leads appear intact without fracture. Mild pulmonary vascular congestion. Small bilateral pleural effusions. No pneumothorax. Partially imaged upper abdomen unremarkable.

Transthoracic Echocardiogram (August 9, 1996, 1015 hours):

Study Quality: Technically adequate despite patient positioning limitations.

Left Ventricle: Moderately dilated (LVEDD 6.1 cm). Severely reduced systolic function with estimated ejection fraction of 30%. Severe hypokinesis of anterior wall, anteroseptum, and apex consistent with prior anteroseptal infarction. Inferior and inferolateral walls demonstrate mild hypokinesis. No left ventricular thrombus visualized on this study.

Right Ventricle: Normal size. Mildly reduced systolic function. TAPSE 1.4 cm.

Left Atrium: Moderately dilated (volume index 38 mL/m²).

Right Atrium: Mildly dilated.

Mitral Valve: Moderate (2-3+) mitral regurgitation with central jet. Valve leaflets appear structurally normal; regurgitation likely functional secondary to LV dilatation and dysfunction. Mean gradient 3 mmHg.

Aortic Valve: Trileaflet. Mild sclerosis without significant stenosis. Trace aortic regurgitation. Peak velocity 1.8 m/s.

Tricuspid Valve: Mild (1+) tricuspid regurgitation. Estimated RVSP 38 mmHg + RAP.

Pulmonic Valve: Normal structure and function.

Pericardium: No effusion.

IVC: Normal caliber with appropriate respiratory variation.

Coronary Angiography (Records from March 1995, obtained from chart review):

Three-vessel coronary artery disease. Left main coronary artery without significant stenosis. Left anterior descending artery with 90% proximal stenosis, status-post drug-eluting stent placement with good angiographic result. Left circumflex artery with diffuse 40-50% disease. Right coronary artery with 70% mid-vessel stenosis, status-post bare metal stent placement. Left ventriculography demonstrated anteroapical akinesis with EF estimated at 35%.

Laboratory Data (August 9, 1996):

Test	Result	Reference Range
Sodium	138 mmol/L	135-145
Potassium	3.6 mmol/L	3.5-5.0
Chloride	104 mmol/L	98-107
Bicarbonate	19 mmol/L	22-28
BUN	42 mg/dL	7-20
Creatinine	1.8 mg/dL	0.7-1.3
Glucose	168 mg/dL	70-100
Calcium	8.9 mg/dL	8.5-10.5
Magnesium	1.9 mg/dL	1.7-2.2
Troponin I	2.8 ng/mL	<0.04
CK-MB	18.4 ng/mL	<5.0
BNP	1240 pg/mL	<100
Lactate	6.2 mmol/L	0.5-2.0
WBC	14.2 × $10^3/\mu L$	4.5-11.0
Hemoglobin	10.8 g/dL	13.5-17.5
Hematocrit	32.1%	41-53
Platelets	188 × $10^3/\mu L$	150-400
PT	12.8 seconds	11.0-13.5
INR	1.1	0.9-1.1
aPTT	28.4 seconds	25-35

Arterial Blood Gas (1000 hours, FiO2 0.60):

Parameter	Value
pH	7.31
pCO2	39 mmHg

pO2	186 mmHg
HCO3	20 mmol/L
Base Excess	-5.2
SaO2	99%
Lactate	4.8 mmol/L

IMPRESSION AND ASSESSMENT

Mr. Hickle134 is an 81-year-old gentleman with significant cardiovascular risk factors and known structural heart disease who presents following witnessed out-of-hospital cardiac arrest with ventricular fibrillation as the presenting rhythm. He achieved return of spontaneous circulation after bystander CPR and appropriate advanced cardiac life support in the field.

Primary Issues:

1. Cardiac Arrest, Ventricular Fibrillation, Status-Post Successful Resuscitation

The patient experienced sudden cardiac arrest this morning with documented ventricular fibrillation. Given his extensive history of coronary artery disease with prior anteroseptal myocardial infarction, severely reduced left ven